KARANVEER SINGH

LinkedIn | GitHub | Medium

PROFESSIONAL PROFILE

Second-year MEng Electrical and Electronics Engineering student with practical experience in PCB design, circuit prototyping, failure analysis and hardware debugging through MBDA and Formula Student. Developing a PCBWay-sponsored robot controller that combines power conversion, motor drive, sensing and embedded interfaces. Interested in autonomous aircraft and mobile robots, rapid prototyping and electronics that progress from schematic to physical test.

TECHNICAL SKILLS

PCB & electronics: KiCad, OrCAD, Altium Designer and ZUKEN; schematic capture, PCB layout, component and footprint selection, power electronics, analogue and digital circuits.

Laboratory: Oscilloscopes, CAN analysers, electronic test equipment, soldering and rework, circuit prototyping, fault investigation and hardware debugging.

Embedded & software: C, C++, Python, MATLAB/Simulink, CAN, UART, I2C, Git and GitHub; Fusion 360 and SolidWorks.

EDUCATION

MEng (Hons) Electrical and Electronics Engineering, University of East Anglia | Norwich   2024 - 2028

- Predicted First-Class Honours; achieved highest grade within cohort in Electrical Machines and Power Electronics module
- Modules: Electrical Machines & Power Electronics, Analogue & Digital Electronics, mathematics, programming and thermofluids.
- Research mentee under Dr Dennis Fitzpatrick, developing PCB-design and electronics-assembly skills through regular laboratory work and technical workshops.

ENGINEERING EXPERIENCE

Electronics Engineering Placement, MBDA | Stevenage   Jun 2026 - Present

- Designed and prototyped circuits for experimental defence electronics, producing a PCB as the final deliverable.
- Investigated potential failure points in customer prototype boards using electronic test equipment and CT inspection; produced a detailed technical report for the customer and Electrical Engineering department.
- Supported PCB routing investigations with senior engineers, considering practical constraints relevant to future experimental hardware.
- Worked alongside experienced electronics engineers in a controlled defence environment, communicating design findings and responding to technical review.

Electrical Systems Lead, UEA Formula Student   2025 - Present

- Lead development of the vehicle's low-voltage electrical architecture and CAN network between the VCU and motor controllers, with attention to grounding and signal integrity in a high-EMI environment.
- Develop shutdown and brake-plausibility circuitry; use oscilloscopes and CAN-analysis tools to diagnose signals and communication frames against Formula Student requirements.
- Coordinate electrical documentation, sensor placement and telemetry integration with mechanical and controls teams.

Nuffield Engineering Placement, Cranfield University | Bedford   Jul 2023 - Aug 2023

- Analysed VTOL aircraft performance across alternative designs, system architectures and propulsion configurations using MATLAB simulation and data analysis.
- Presented findings and engineering trade-offs to peers and a Cranfield University review panel.

SELECTED ENGINEERING PROJECTS

Advanced 2WD Robot Controller | PCBWay-sponsored 2026 - Present

• Developing a custom KiCad controller PCB for a 2S LiPo mobile robot, integrating regulated 6 V motor and 5 V logic rails, dual-motor control, current sensing, battery monitoring and encoder inputs.

• Integrated UART and I2C expansion for Bluetooth, ultrasonic sensing and future peripherals, with protection and testability considered at system level.

• Selected components and footprints from primary datasheets and iterated the design through ERC/DRC, pin-mapping, grounding, bootstrap, protection and manufacturability reviews before fabrication.

• Documenting the design rationale and technical progress as part of a PCBWay-sponsored public project series.

Autonomous Drone Systems | GitHub & Medium 2025

• Modelled mission profiles, battery energy, power demand and flight-time trade-offs for autonomous multirotor systems.

• Implemented KNN-based control logic and compared system configurations to improve endurance and mission performance.

• Documented assumptions, modelling methods and results through public technical writing and version-controlled code.

Amplitude-Modulation Audio Communication Nov 2025

• Designed and simulated an AM transmitter-receiver chain using MATLAB/Simulink and Arduino-based signal generation, including modulation, channel noise, filtering and demodulation.

• Applied phasor and time/frequency-domain analysis; calculated gain, modulation index, bandwidth, DC offset and power relationships.

• Ran parameter sweeps to evaluate how modulation index and noise affected the recovered output.

LEADERSHIP & TECHNICAL ENGAGEMENT

Co-founder & Electrical Lead, UEA Innovators 2024 - Present

• Lead electronics activity for a project converting a Volkswagen Polo into an autonomous research vehicle and coordinate multidisciplinary technical work.

• Organise practical engineering workshops for 50+ members, including PCB-design and electronics sessions delivered with university academics.

Health and Safety Officer, UEA Engineering Society 2024 - Present

• Produce risk assessments and support safe delivery of technical workshops, competitions and laboratory activities.

ADDITIONAL PROJECTS, EMPLOYMENT & ACHIEVEMENTS

Software and AI: C Mega Calculator; HealthBridge Python/LLaMA healthcare chatbot; explainable-AI dementia classifier using hyperparameter tuning, SHAP and LIME.

Technical communication: Publish first-principles engineering articles on Medium and document project decisions, modelling methods and results for external review, attracting +3000 readers in 3 months.

Employment: Crew Member, McDonald’s, Bedford (Jul 2023 - Jan 2024); worked accurately under time pressure while handling customer orders and coordinating with colleagues.

Awards: Impetus Education Award for academic excellence (2024); CyberCenturion National Semi-Finalist (2022); 2nd place, Pygame Community Jam (2022).